

# Probabilistic Machine Learning

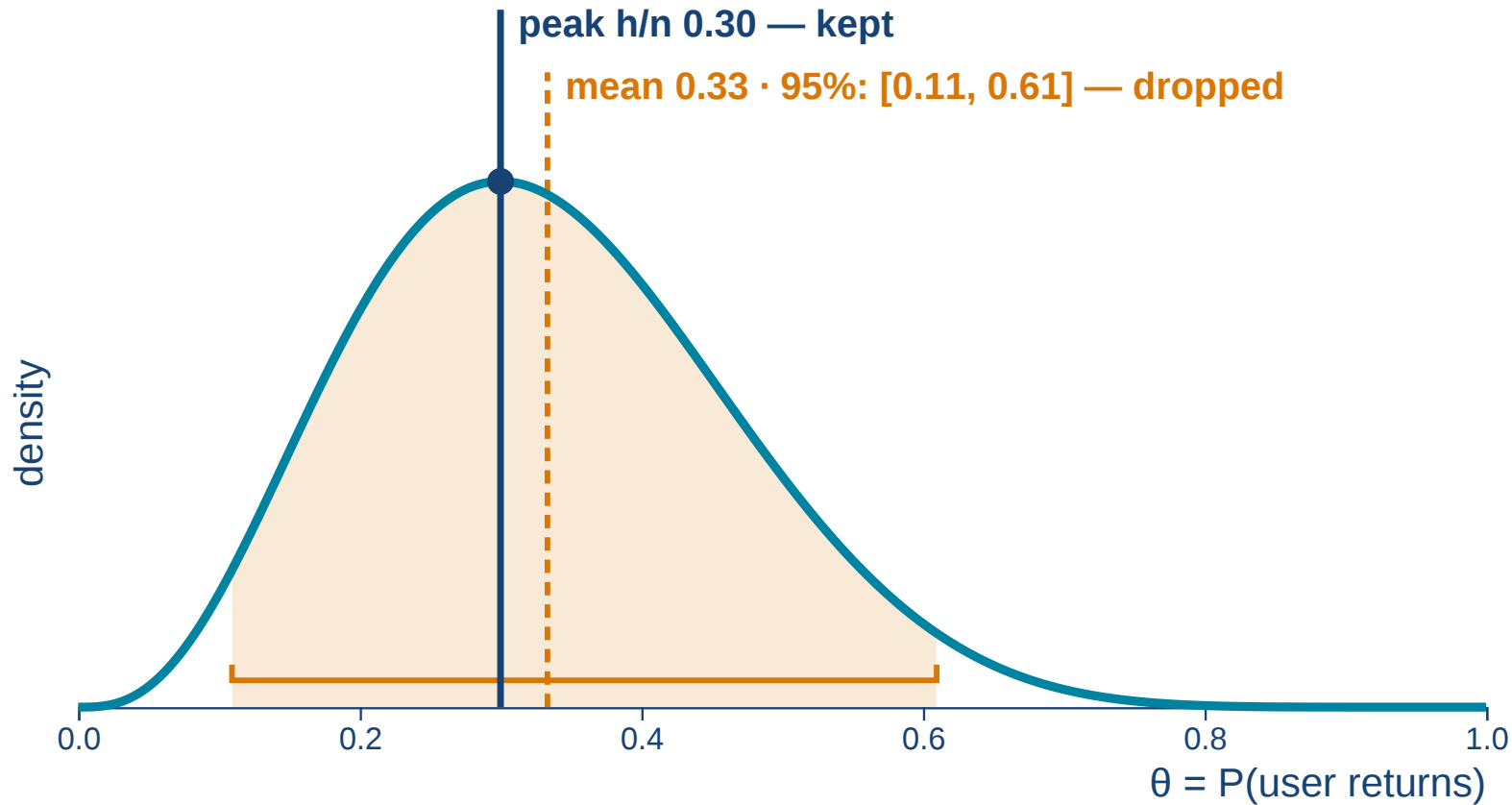
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## Lecture 03: Linear Models — Keep the Peak, Model Only $p(y | x)$

Prof. Dr. Michael Granitzer

# Where We Left Off

Lecture 02: 3 of 10 users, flat prior  $\rightarrow$  Beta(4, 8)



One posterior, three readings:

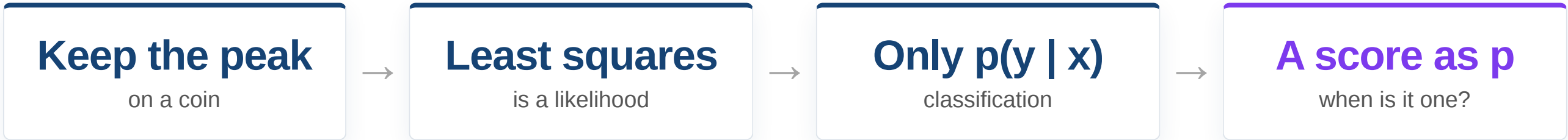
- **peak**  $0.30 = h/n$
- **mean** 0.33
- **95% interval** [0.11, 0.61]

Today keeps only the **peak** — and you already know the methods that do.

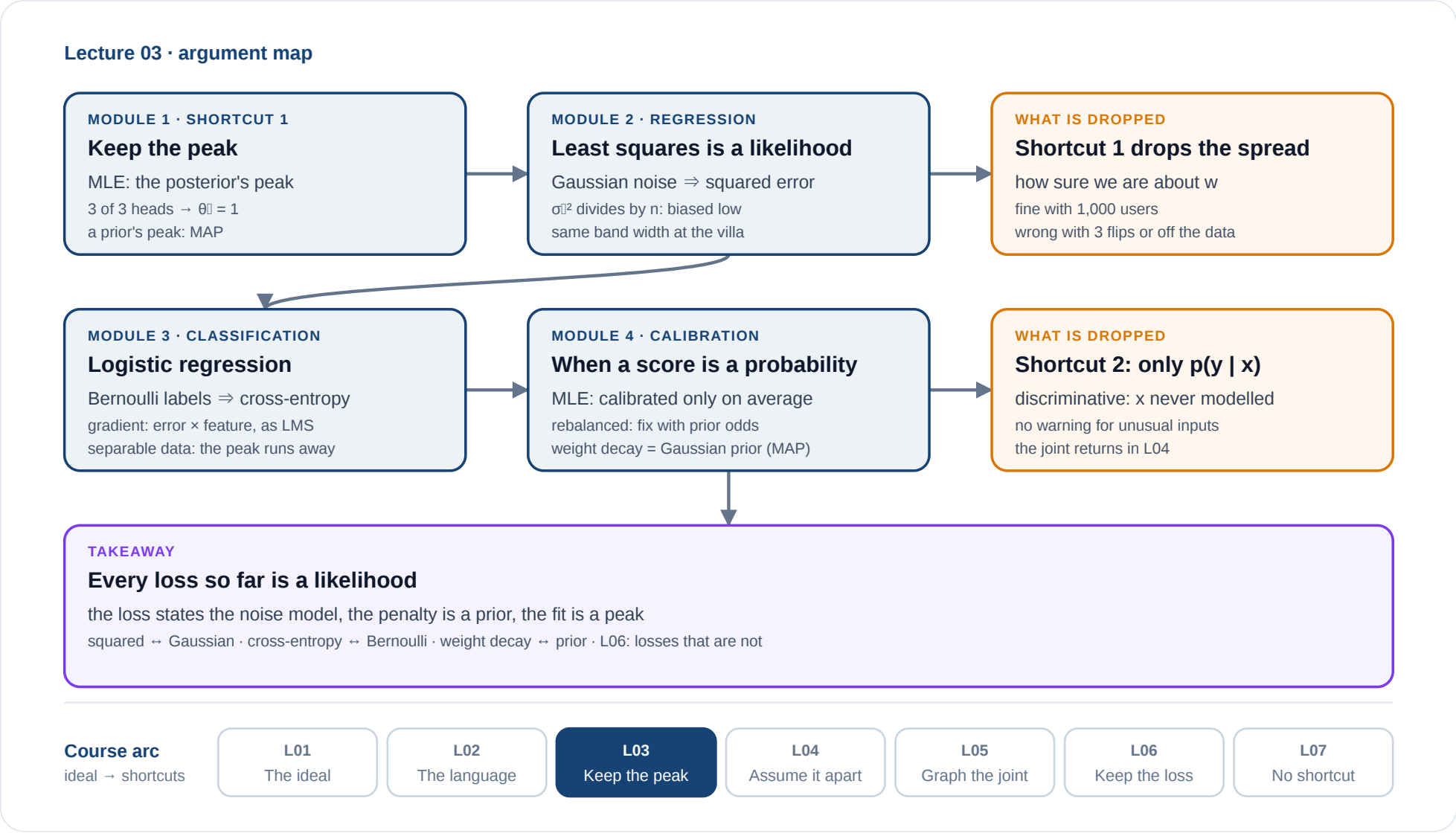
# Today: Two Shortcuts

Lecture 01's table, rows 1 and 2 — methods you have used, read for what they drop:

| shortcut           | instead of the ideal                                                        | what is lost             | where               |
|--------------------|-----------------------------------------------------------------------------|--------------------------|---------------------|
| Keep the peak      | the most probable parameter — MLE or MAP                                    | how sure we are about it | a coin · regression |
| Only $p(y \mid x)$ | model the label given the features, never the joint — <i>discriminative</i> | every question about $x$ | classification      |



# Argument Map



# Keep the Peak

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**Guiding Question:** Last week's posterior gave a peak, a mean and an interval. What do we lose when we keep only the peak — and why does almost everyone do it?

# The Likelihood, Read Along the Other Axis

The **likelihood** is the probability of the observed data, read as a function of the **parameter** with the data held fixed:  $L(\theta) = p(D \mid \theta)$ . The **maximum likelihood estimate** is its peak,  $\hat{\theta}_{\text{ML}} = \arg \max_{\theta} p(D \mid \theta)$ .

## Lecture 02, 3 of 10 users

$$p(\theta \mid D) \propto \theta^3(1 - \theta)^7 \cdot 1$$

With a **flat prior** the posterior *is* the normalised likelihood — so its peak  $h/n = 0.3$  is the maximum likelihood estimate.

## Not a distribution over $\theta$

$$\int_0^1 \theta^3(1 - \theta)^7 d\theta = \frac{1}{1320}$$

The likelihood is a probability over **data**. Read along  $\theta$  it need not integrate to 1 — only the posterior does.

# Deriving $h/n$ Without a Posterior

Lecture 02's i.i.d. assumption lets the likelihood factorise over rows; the logarithm turns the product into a sum without moving the peak:

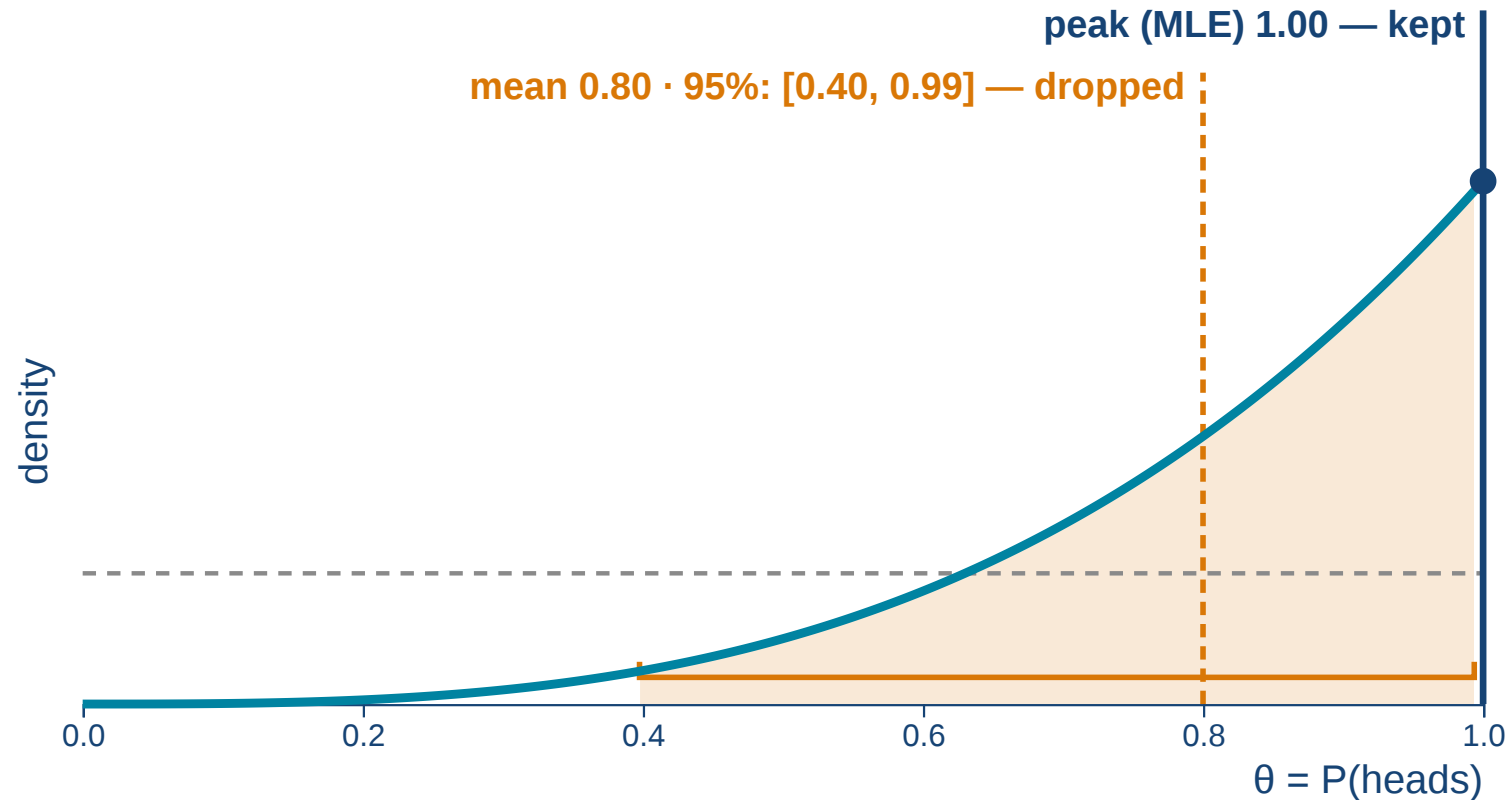
$$\ell(\theta) = \log \prod_{i=1}^n \theta^{y_i} (1 - \theta)^{1-y_i} = h \log \theta + (n - h) \log(1 - \theta)$$
$$\frac{d\ell}{d\theta} = \frac{h}{\theta} - \frac{n - h}{1 - \theta} = 0 \quad \implies \quad \hat{\theta}_{\text{ML}} = \frac{h}{n}$$

Last week's peak, found by **setting a derivative to zero** — no prior, no normaliser, no integral. The recipe works unchanged for a million parameters, where nobody can draw the posterior.

# The Peak Can Lie

Three flips, **three heads**. Maximum likelihood:  $\hat{\theta}_{\text{ML}} = 3/3 = 1$ .

Flat prior (dashed), 3 heads: posterior Beta(4, 1)



Predict before the second click: where do the posterior mean and the 95% interval lie?

# A Prior's Peak: MAP

Keep the prior, and still take only the peak — the **maximum a posteriori** estimate. With a  $\text{Beta}(a, b)$  prior, the same derivative gives

$$\hat{\theta}_{\text{MAP}} = \arg \max_{\theta} p(D \mid \theta)p(\theta) = \frac{h + a - 1}{n + a + b - 2}$$

| prior           | data                 | MLE $h/n$ | MAP         | posterior mean |
|-----------------|----------------------|-----------|-------------|----------------|
| Beta(1, 1) flat | 3 of 3               | 1         | 1           | 0.80           |
| Beta(2, 2)      | 3 of 3               | 1         | <b>0.80</b> | 0.71           |
| Beta(2, 2)      | 8 of 10 (Lecture 02) | 0.80      | <b>0.75</b> | 0.71           |

The prior shifts the peak as if  $a - 1$  yeses and  $b - 1$  noes had been seen before the data — a **flat** prior shifts nothing. MAP pulls 3 of 3 back to 0.8 — and still reports one number.

# Why Everyone Keeps the Peak

|                | the whole posterior                               | only the peak     |
|----------------|---------------------------------------------------|-------------------|
| needs          | the normaliser $p(D)$ — an integral over $\theta$ | a derivative      |
| in 1 dimension | a Beta table                                      | $h/n$             |
| in a million   | an integral nobody can compute exactly            | gradient descent  |
| costs          | computation                                       | the <b>spread</b> |

With **1,000 users** the posterior is Beta(301, 701): standard deviation 0.015. The peak is nearly everything there is to know.

# Mini-Exercise

Two analysts both report  $\hat{\theta} = 0.3$ . One saw **3 of 10** users return, the other **300 of 1,000**.

What does the reported number hide — and which estimate would you bet on?

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The **spread**. Lecture 02's posteriors:  $[0.11, 0.61]$  after 10 users,  $[0.27, 0.33]$  after 1,000 — the same peak, an interval nine times wider. Bet on the second; a point estimate cannot tell you to.

# Takeaway

**Shortcut 1 — keep the peak.** Maximum likelihood (flat prior) and MAP (with a prior) keep the posterior's most probable point. It is cheap — a derivative instead of an integral — and harmless with plenty of data. It drops the **spread**: how sure we are about the parameter.

# Least Squares Is a Likelihood

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**Guiding Question:** You have minimised squared error many times. What did that choice assume about the world — and what does it leave out?

# Linear Regression in One Slide

A price model for flats of **40–120 m<sup>2</sup>**, with  $x_0 = 1$  for the intercept — everything else is in the backup:

| piece        | recap                                                                                                                       |
|--------------|-----------------------------------------------------------------------------------------------------------------------------|
| model        | $\hat{y} = \mathbf{w}^\top \mathbf{x} = w_0 + w_1 x_1 + \cdots + w_d x_d$                                                   |
| objective    | $\text{RSS}(\mathbf{w}) = \sum_{i=1}^n (y_i - \mathbf{w}^\top \mathbf{x}_i)^2$                                              |
| closed form  | normal equations: $\hat{\mathbf{w}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$                         |
| iterative    | LMS, one example at a time: $\mathbf{w} \leftarrow \mathbf{w} + \eta(y_i - \hat{y}_i)\mathbf{x}_i$ — error $\times$ feature |
| justified by | Gauss–Markov: best <i>linear unbiased</i> estimator if the errors have mean zero, equal variance and no correlation         |

# The Same Model, Told Differently

## Classical account

Choose the line that **minimises squared error**:

$$\hat{\mathbf{w}} = \arg \min_{\mathbf{w}} \sum_i (y_i - \mathbf{w}^\top \mathbf{x}_i)^2$$

## Probabilistic account

State how a price **arises** for a given flat:

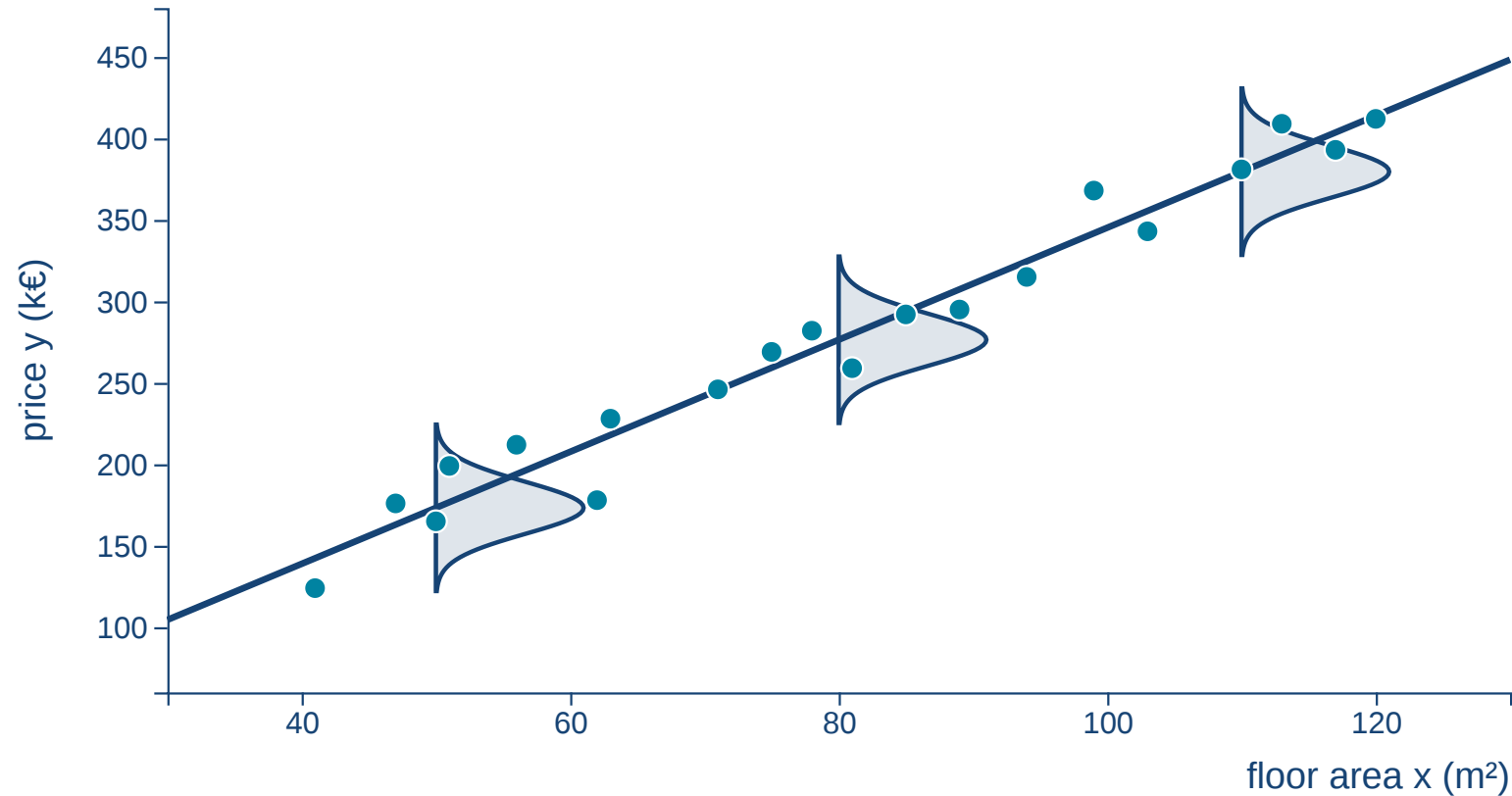
$$y = \mathbf{w}^\top \mathbf{x} + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, \sigma^2)$$

then keep the peak of the likelihood.

Do the two accounts give the same  $\hat{\mathbf{w}}$  — and if so, what did the classical account assume without saying it?

# A Distribution at Every $x$

$p(y | x) = \mathcal{N}(\hat{w}_0 + \hat{w}_1 x, \sigma^2)$ ,  $\sigma = 16 \text{ k€}$  — the same bell at every  $x$



A **conditional distribution**  $p(y | x) = \mathcal{N}(\mathbf{w}^\top \mathbf{x}, \sigma^2)$ : one bell per  $x$ , **equally wide**. Nothing about how  $x$  arises.

# The Derivation

With  $p(y_i \mid \mathbf{x}_i, \mathbf{w}) = \mathcal{N}(y_i \mid \mathbf{w}^\top \mathbf{x}_i, \sigma^2)$  and i.i.d. rows:

$$\ell(\mathbf{w}) = \sum_{i=1}^n \log \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left( -\frac{(y_i - \mathbf{w}^\top \mathbf{x}_i)^2}{2\sigma^2} \right) = \underbrace{-\frac{n}{2} \log(2\pi\sigma^2)}_{\text{no } \mathbf{w}} - \frac{1}{2\sigma^2} \text{RSS}(\mathbf{w})$$

Maximising the Gaussian log-likelihood **is** minimising RSS:  $\hat{\mathbf{w}}_{\text{ML}} = \hat{\mathbf{w}}_{\text{OLS}}$ . Squared error is the exponent of the bell.

# What Squared Error Assumed

| the noise model says       | squared error silently assumes    | check on the residuals       |
|----------------------------|-----------------------------------|------------------------------|
| $\varepsilon$ has mean 0   | the line is the right shape       | no pattern against $x$       |
| one $\sigma$ for every $x$ | <b>homoscedastic</b> noise        | no fan shape                 |
| rows independent           | i.i.d. (Lecture 02)               | no pattern in time or space  |
| $\varepsilon$ Gaussian     | thin tails: large errors are rare | no heavy tails, few outliers |

Choosing a loss and choosing a noise model are **one decision**. Squared error  $\Leftrightarrow$  Gaussian noise; absolute error  $\Leftrightarrow$  Laplace noise — heavier tails, so a large residual is less surprising and an outlier pulls the fit less.

# The Peak of $\sigma^2$

The same likelihood also has a peak in  $\sigma^2$ :

$$\frac{\partial \ell}{\partial \sigma^2} = -\frac{n}{2\sigma^2} + \frac{\text{RSS}}{2\sigma^4} = 0 \quad \implies \quad \hat{\sigma}_{\text{ML}}^2 = \frac{\text{RSS}}{n}$$

Which of Lecture 02's two variances is this — and is it too small or too large?

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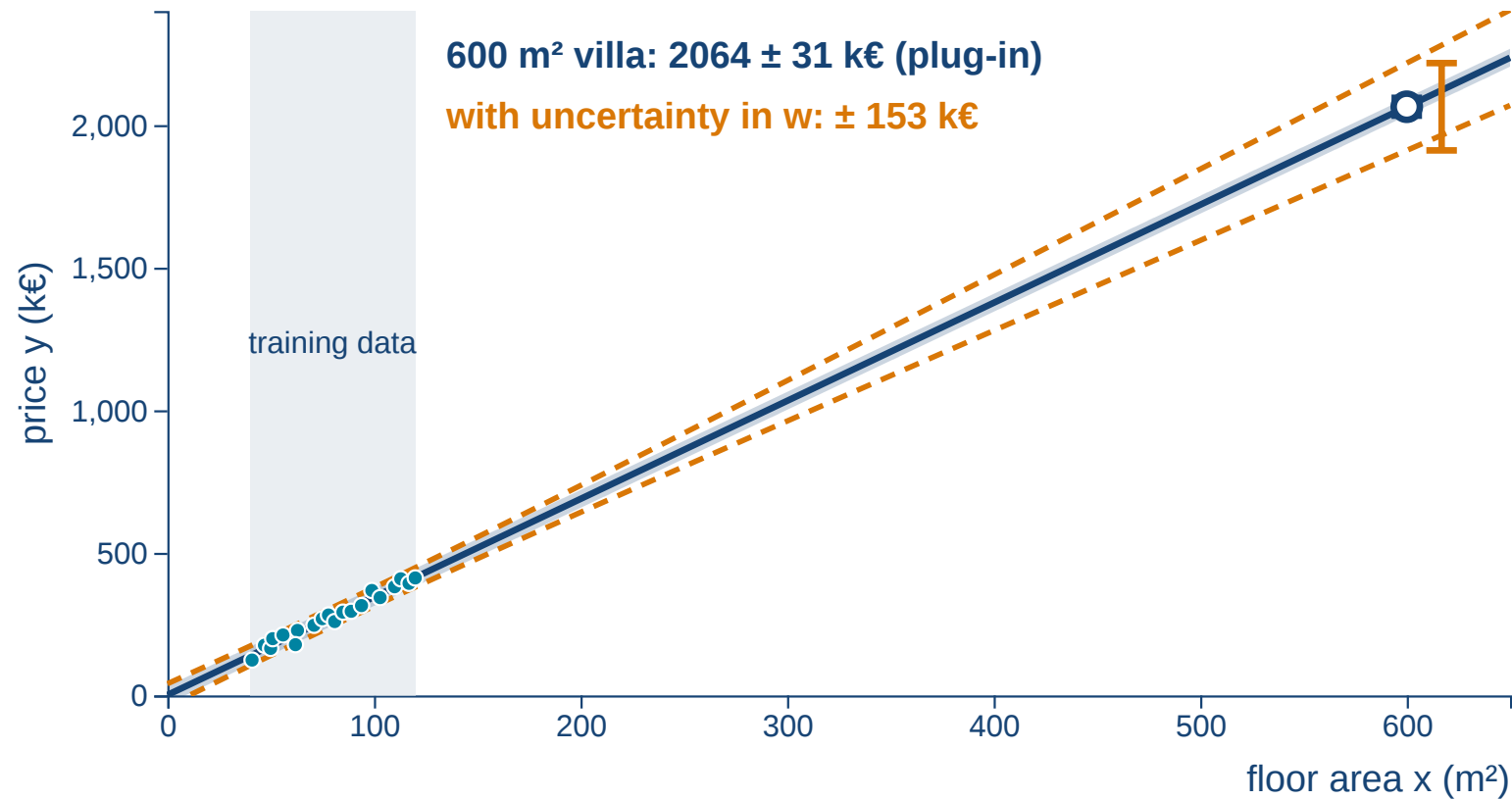
Which of Lecture 02's two variances is this — and is it too small or too large?

The  $1/n$  one. The line was fitted to the same data, so residuals are smaller than the true errors:  $\hat{\sigma}_{\text{ML}}^2$  is **biased low**. The unbiased version divides by  $n - p$  for  $p$  fitted weights.

Houses:  $n = 20$ ,  $p = 2$ ,  $\text{RSS} \approx 5,097 \rightarrow \hat{\sigma} \approx 16.0$  instead of 16.8 k€.

# The Villa Again

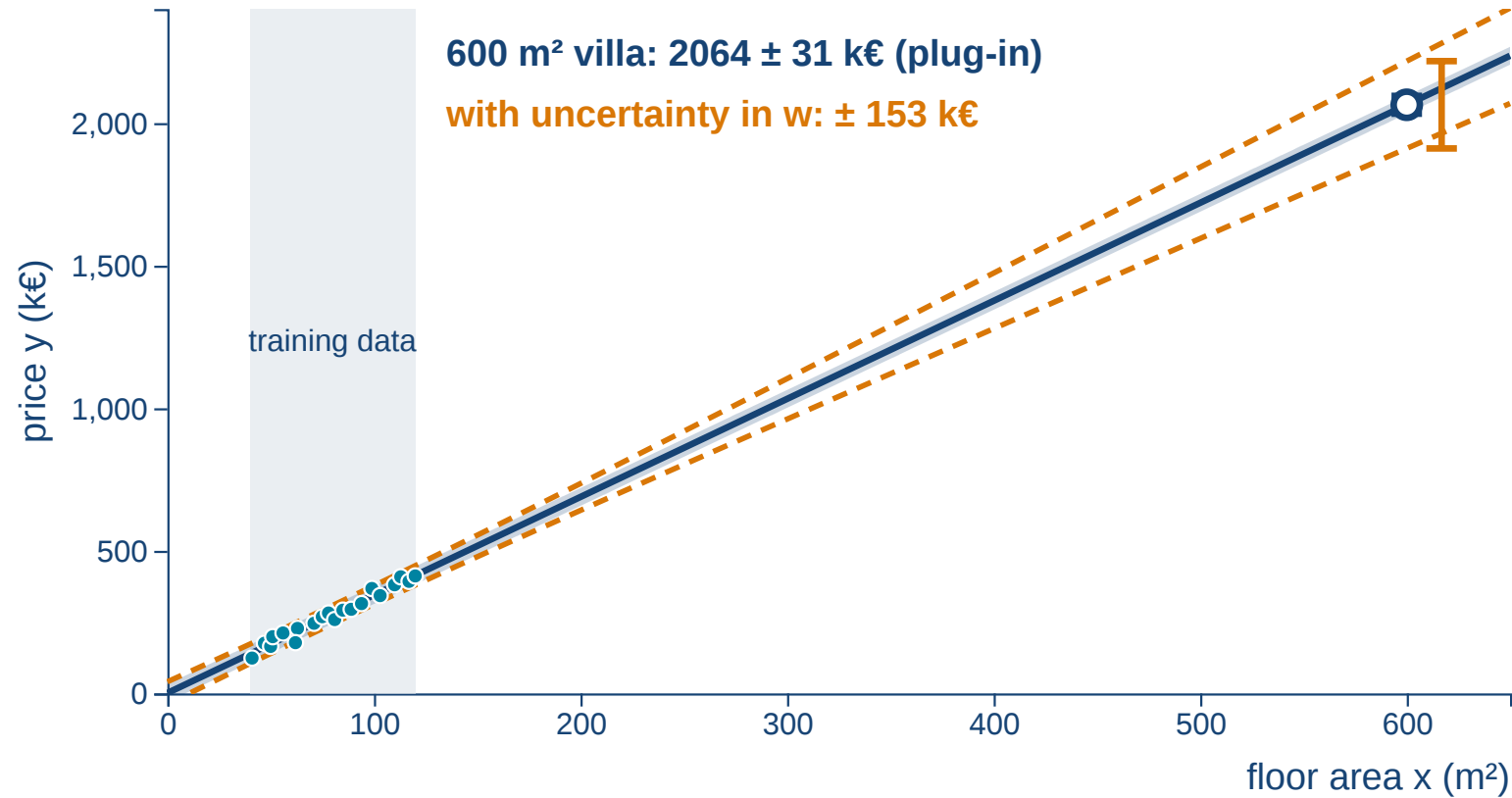
The peak dropped the uncertainty about  $w$  — it grows off the data



The band is as narrow at the villa as at a flat. Is the model as sure?

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The band is as narrow at the villa as at a flat. Is the model as sure?

No:  $\pm 31$  k€ is only the noise  $\hat{\sigma}$ . The uncertainty about  $w$  — dropped with the peak — grows off the data:  $\pm 153$  k€ at the villa (Lecture 07).

# Mini-Exercise

The residuals of your house-price model fan out: about  $\pm 10$  k€ for 40 m<sup>2</sup> flats,  $\pm 60$  k€ for 120 m<sup>2</sup> flats.

Which assumption fails — and which output of the model can you no longer trust?

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**One  $\sigma$  for every  $x$**  (homoscedasticity). The line may still be fine, but the **band** is not: a single  $\hat{\sigma}$  is too wide for small flats and too narrow for large ones.

# Takeaway

Least squares **is** maximum likelihood under Gaussian noise: the loss states the noise model, and the model is a distribution  $p(y \mid x)$ . It is also a **peak**:  $\hat{\sigma}^2$  divides by  $n$ , and the band is as narrow at the villa as at a flat — the uncertainty about  $\mathbf{w}$  was dropped.

# Classification: Model Only $p(y | x)$

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**Guiding Question:** Lecture 01's fraud model said 0.55. Where does such a number come from — and what does the model refuse to know?

# Why Not Regress on 0/1?

Code fraud as  $y = 1$ , legitimate as  $y = 0$ , and fit Module 2's line.

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Code fraud as  $y = 1$ , legitimate as  $y = 0$ , and fit Module 2's line.

## Squared error punishes being right

A transaction far on the correct side has a large residual, so least squares drags the boundary towards cases it already classifies correctly.

## The output is not a probability

$\mathbf{w}^\top \mathbf{x}$  can be 17 or  $-40$ . The Gaussian bell of Module 2 puts mass on every real  $y$  — for a label that is 0 or 1.

Keep the linear score, change what it models: the **log-odds**, and a noise model that produces 0s and 1s.

# Log-Odds Are Linear

$$\ln \frac{P(y = 1 \mid \mathbf{x})}{1 - P(y = 1 \mid \mathbf{x})} = \mathbf{w}^\top \mathbf{x} \quad \Longleftrightarrow \quad P(y = 1 \mid \mathbf{x}) = \sigma(\mathbf{w}^\top \mathbf{x}) = \frac{1}{1 + e^{-\mathbf{w}^\top \mathbf{x}}}$$

## read the weights as

the intercept: the log-odds when every feature is 0

a weight: what one unit of a feature **adds** to the log-odds

the boundary: where  $\mathbf{w}^\top \mathbf{x} = 0$

## fraud example

no features at all:  $w_0 = \ln \frac{1}{999} \approx -6.9$ , the base rate 1 in 1,000

"new device" with  $w = \ln 6 \approx 1.8$ : odds  $\times 6$

score 0.5 — still a straight line

# Where Does Cross-Entropy Come From?

$$L(\mathbf{w}) = - \sum_{i=1}^n \left[ y_i \ln h_i + (1 - y_i) \ln(1 - h_i) \right], \quad h_i = \sigma(\mathbf{w}^\top \mathbf{x}_i)$$

You have minimised this loss many times. Which noise model does it state?

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Given  $\mathbf{x}$ , the label is a **Bernoulli** draw with bias  $h$ :  $P(y_i \mid \mathbf{x}_i, \mathbf{w}) = h_i^{y_i} (1 - h_i)^{1-y_i}$ . With i.i.d. rows,

$$\ell(\mathbf{w}) = \sum_{i=1}^n \left[ y_i \ln h_i + (1 - y_i) \ln(1 - h_i) \right] = -L(\mathbf{w})$$

Cross-entropy **is** the Bernoulli negative log-likelihood — Module 1's coin, biased by  $\mathbf{x}$ . How  $\mathbf{x}$  arises is never said.

# The Same Update Again

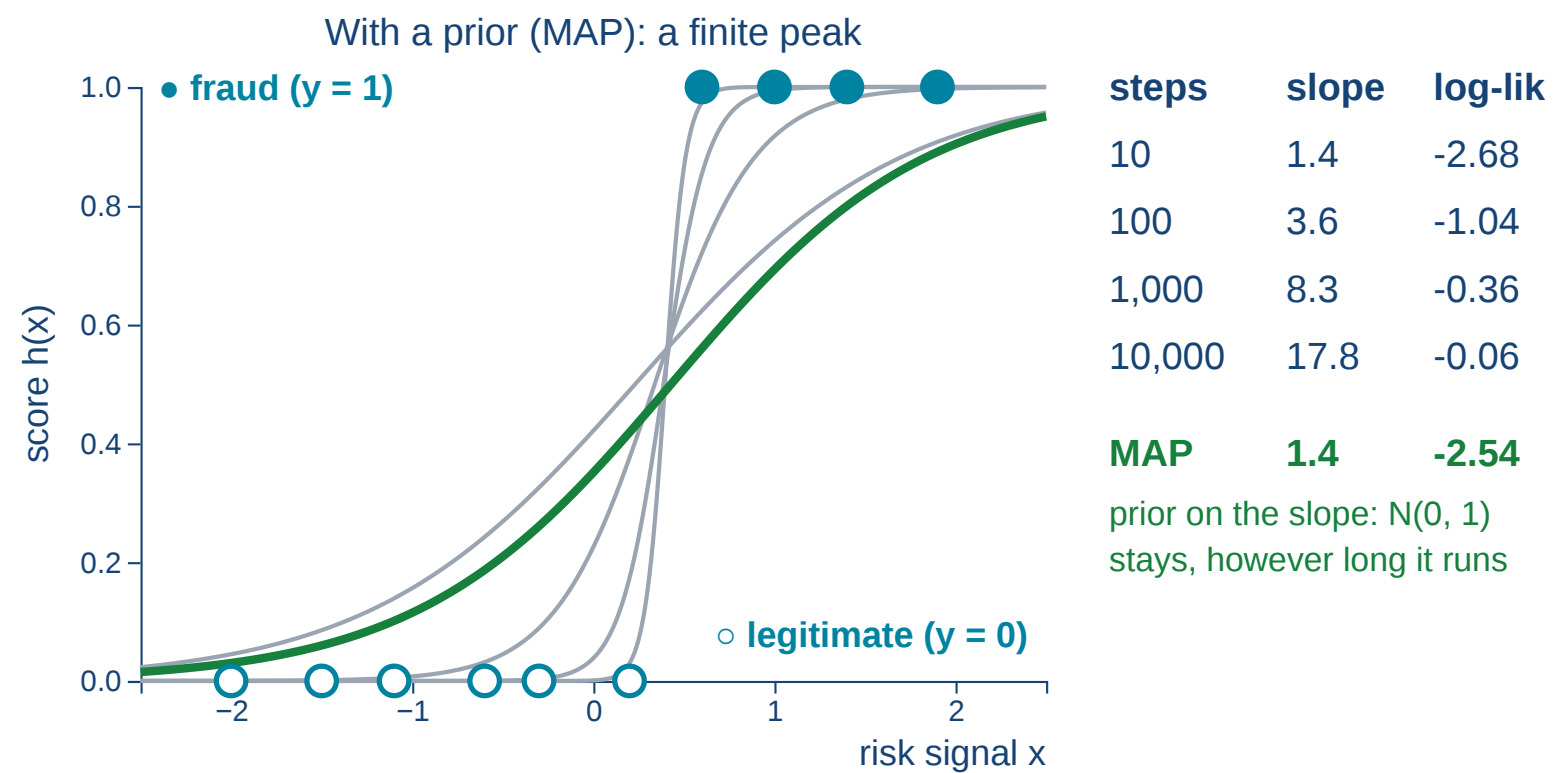
With  $\sigma' = \sigma(1 - \sigma)$ , the gradient of the log-likelihood is

$$\frac{\partial \ell}{\partial \mathbf{w}} = \sum_{i=1}^n (y_i - h_i) \mathbf{x}_i$$

| model               | noise     | loss          | gradient of $\ell$                                                            |
|---------------------|-----------|---------------|-------------------------------------------------------------------------------|
| linear regression   | Gaussian  | squared error | $\frac{1}{\sigma^2} \sum_i (y_i - \mathbf{w}^\top \mathbf{x}_i) \mathbf{x}_i$ |
| logistic regression | Bernoulli | cross-entropy | $\sum_i (y_i - \sigma(\mathbf{w}^\top \mathbf{x}_i)) \mathbf{x}_i$            |

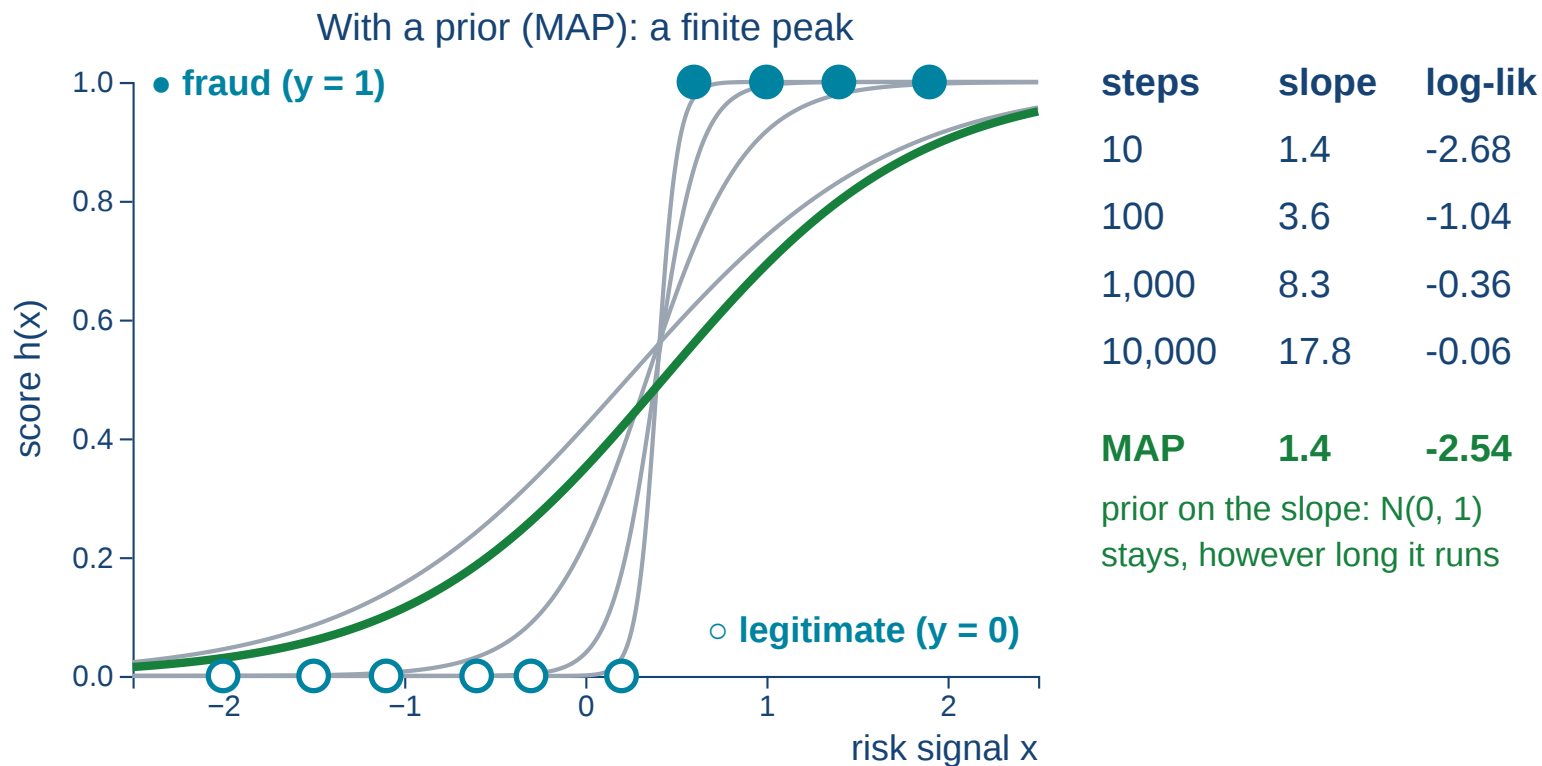
Error  $\times$  feature — the LMS update with a sigmoid in the prediction; no closed form, so iterate. A fraud scored 0.01 costs  $-\ln 0.01 \approx 4.6$ , scored 0.99 only 0.01: confident mistakes dominate.

# When the Peak Runs Away



Every fraud scores above every legitimate case. Where is the peak of the likelihood?

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**Nowhere.** The likelihood keeps rising as the slope grows; the scores harden to 0 and 1 — three heads in three flips. A prior's peak (MAP) is finite.



# Shortcut 2: Model Only $p(y \mid \mathbf{x})$

Linear and logistic regression are **discriminative**: a model of  $y$  given  $\mathbf{x}$ , none of  $\mathbf{x}$ .

## What is dropped: $p(\mathbf{x})$

- a transaction unlike any seen still gets a confident score
- a missing field cannot be averaged out — it must be filled in
- no way to simulate a plausible transaction

## What is gained

- no assumptions about how features are distributed — where, with hundreds of features, most modelling errors hide
- every parameter serves the decision

Lecture 04 takes the other route: model the joint  $p(\mathbf{x}, y)$  and condition — **generative**.

# Mini-Exercise

Your fraud model scores **0.97** for a transfer 50 times larger than any in its training data.

Should you trust the 0.97 — and could the model itself warn you?

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Should you trust the 0.97 — and could the model itself warn you?

Not blindly: the linear log-odds simply **extrapolate**, like the line at the villa. And no — the model has **no**  $p(\mathbf{x})$ , so nothing in it can see that the amount is unusual. That needs a model of  $\mathbf{x}$  or a separate check.

# Takeaway

Logistic regression is Module 2 with Bernoulli noise: linear log-odds, cross-entropy as the negative log-likelihood, error  $\times$  feature as the gradient. It **keeps the peak** — which runs away on separable data — and **models only**  $p(y \mid \mathbf{x})$ : it never learns what a normal transaction looks like.

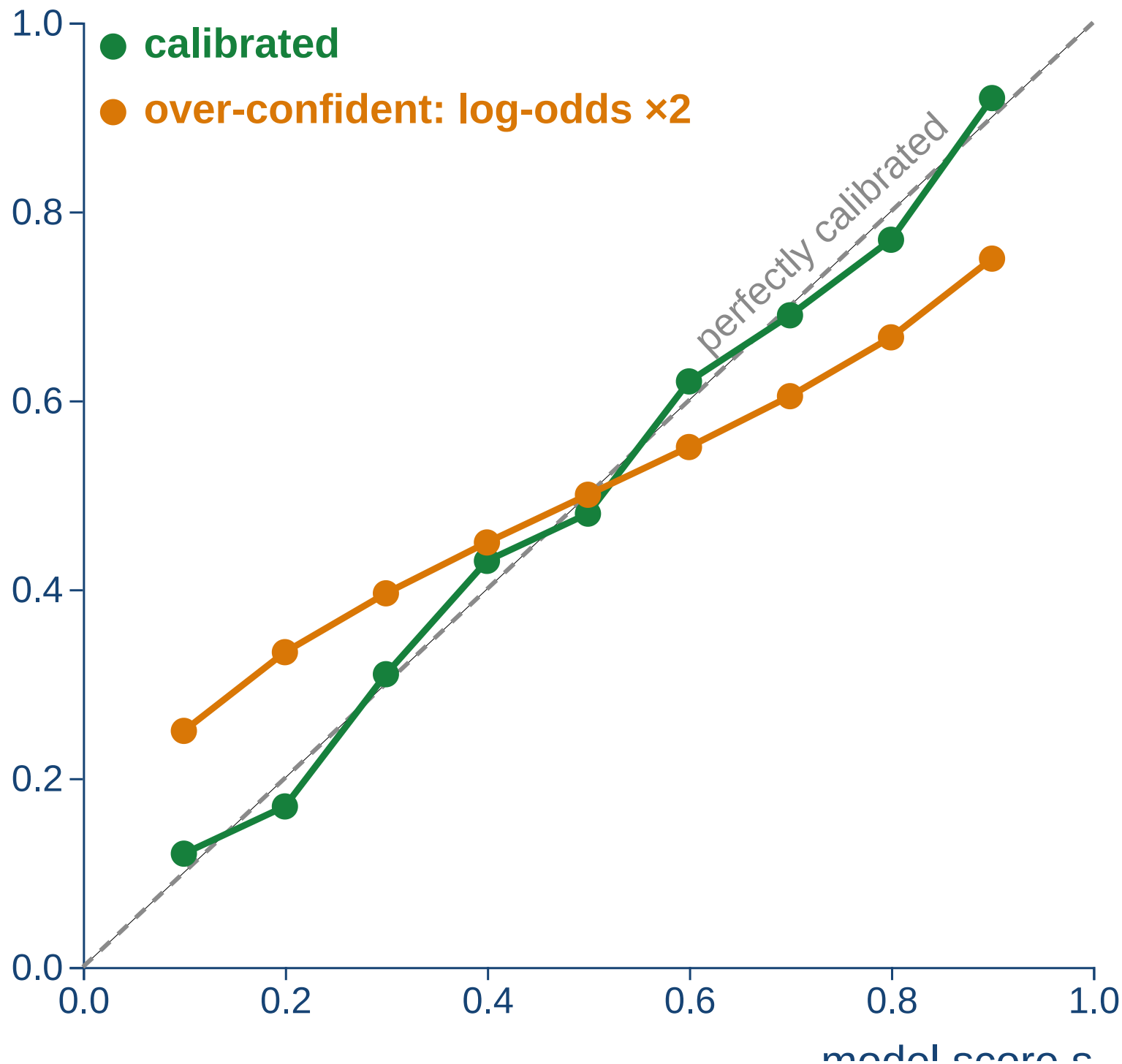
# When Is a Score a Probability?

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**Guiding Question:** Lecture 01 promised to say when a model's output earns the name "probability". When does it — and what breaks it?

# When Is 0.7 a Probability?





A score is **calibrated** if, among all cases scored  $s$ , a fraction  $s$  is positive. A **reliability diagram** plots that fraction against  $s$ .

**Predict before click 2:** a model whose log-odds are twice too large — above or

# What Maximum Likelihood Guarantees

At the peak, the gradient for the intercept ( $x_0 = 1$ ) is zero:

$$\sum_{i=1}^n (y_i - h_i) = 0 \quad \implies \quad \frac{1}{n} \sum_{i=1}^n h_i = \frac{1}{n} \sum_{i=1}^n y_i$$

What does this equation promise — and what does it not?

# What Maximum Likelihood Guarantees

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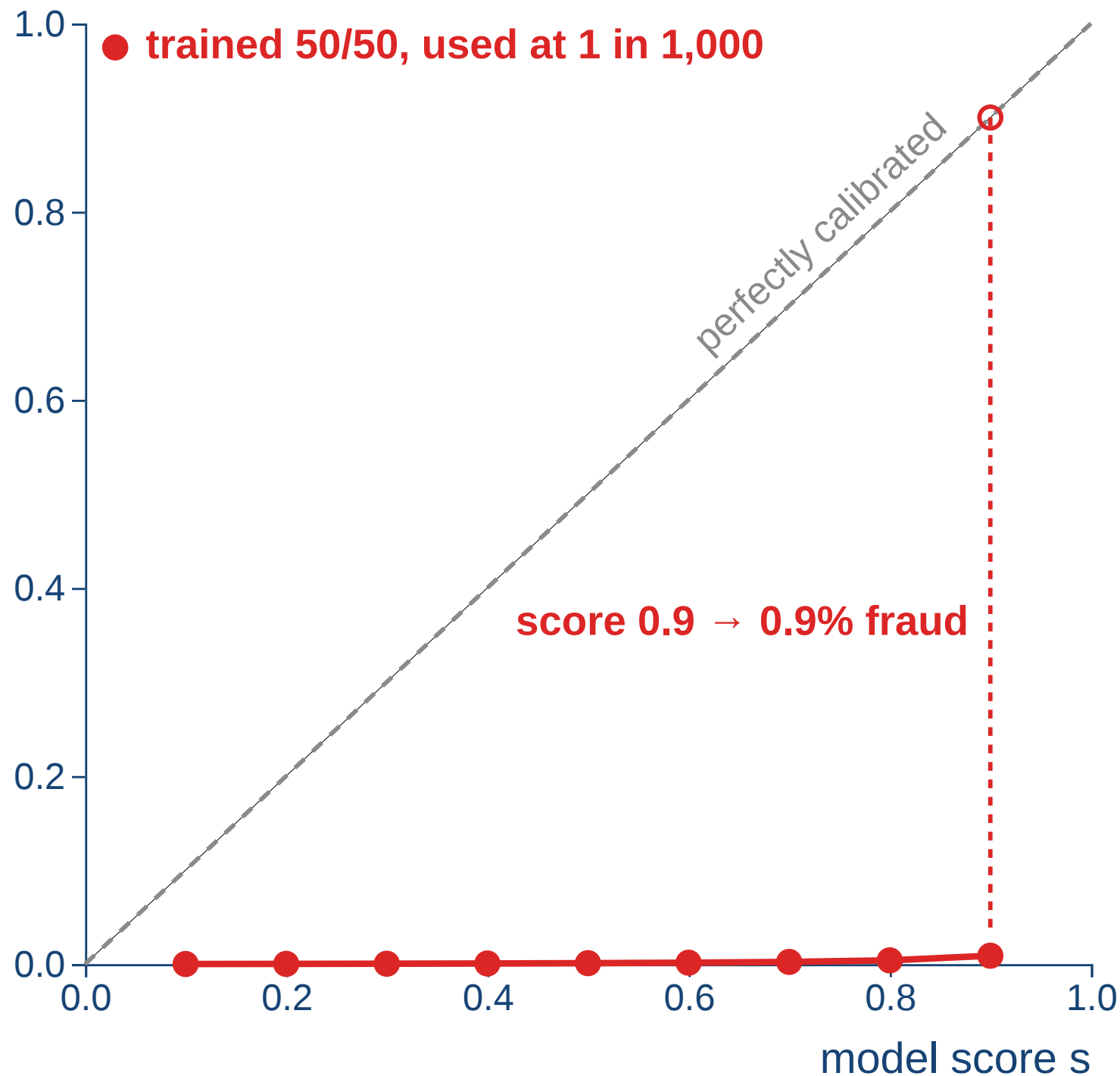
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What does this equation promise — and what does it not?

| promised                                                                    | not promised                                                                      |
|-----------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| <b>average</b> score = training fraud rate: <i>calibration in the large</i> | each score right — needs log-odds truly linear in $\mathbf{x}$                    |
| on the <b>training</b> distribution                                         | after a base-rate shift, or <b>rebalancing</b>                                    |
|                                                                             | that it <b>separates</b> : the base rate for everyone is calibrated — and useless |

# Rebalanced Training





Fraud is 1 in 1,000, so practitioners **undersample** legitimate cases to 50/50. The model learns odds for a world where fraud is common.

Bayes in odds form:

$$\text{real odds} = \text{model odds} \times \frac{1/999}{1}$$

Score 0.9: odds 9 → 9/999, about **0.9%**.

# Regularisation Is a Prior

$$\min_{\mathbf{w}} \text{RSS}(\mathbf{w}) + \lambda \|\mathbf{w}\|^2$$

You know this as ridge regression or weight decay. Which prior's peak is it?

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You know this as ridge regression or weight decay. Which prior's peak is it?

MAP from Module 1 with  $\mathbf{w} \sim \mathcal{N}(0, \tau^2 \mathbf{I})$ :  $\log p(\mathbf{w} \mid D) = -\frac{\text{RSS}}{2\sigma^2} - \frac{\|\mathbf{w}\|^2}{2\tau^2} + \text{const}$ , so  $\lambda = \sigma^2 / \tau^2$ .

| penalty                    | prior on $\mathbf{w}$ | what its peak does                                                                       |
|----------------------------|-----------------------|------------------------------------------------------------------------------------------|
| $\lambda \ \mathbf{w}\ ^2$ | Gaussian              | shrinks; keeps separable data finite                                                     |
| $\lambda \ \mathbf{w}\ _1$ | Laplace               | exact zeros — a property of the <b>peak</b> (the kink at 0); the posterior mean has none |

# Mini-Exercise

A fraud model trained on a **50/50** sample is used on real traffic, where fraud is **1 in 1,000**. It outputs **0.5** and **0.8**.

What are the real fraud probabilities?

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What are the real fraud probabilities?

Multiply the odds by  $1/999$ . Score 0.5: odds 1  $\rightarrow$   $1/999$ , about **0.1%** — the base rate itself. Score 0.8: odds 4  $\rightarrow$   $4/999$ , about **0.4%**.

# Takeaway

A score is a probability when it is **calibrated** — checkable on a reliability diagram, not by accuracy. Maximum likelihood guarantees it only on average and only on the training distribution; **rebalancing** breaks it, and the prior odds repair it. **Regularisation is a prior**, and a penalised fit is a MAP estimate.

# Every Loss So Far Is a Likelihood

So far — Lecture 06 meets losses that are not.

| you minimised               | it is $-\log$ of                                                | the fit is           |
|-----------------------------|-----------------------------------------------------------------|----------------------|
| squared error               | Gaussian noise around $\mathbf{w}^\top \mathbf{x}$              | the peak (MLE)       |
| cross-entropy               | Bernoulli labels with bias $\sigma(\mathbf{w}^\top \mathbf{x})$ | the peak (MLE)       |
| $+\lambda \ \mathbf{w}\ ^2$ | a Gaussian prior on $\mathbf{w}$                                | a prior's peak (MAP) |
| $+\lambda \ \mathbf{w}\ _1$ | a Laplace prior on $\mathbf{w}$                                 | a prior's peak (MAP) |

## Dropped: the spread

how sure we are about  $\mathbf{w}$  — **Lecture 07**  
keeps it

## Dropped: $p(\mathbf{x})$

what a normal input looks like — **Lecture 04**  
models it

# Backup: Classical Regression

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For questions: the closed form,  $R^2$ , basis functions, Gauss–Markov and gradient descent  
— what students know from earlier courses.

# Classification versus Regression

## Classification

$Y$  is a finite set of labels.

*Will this customer default — yes or no?*

## Regression

$Y = \mathbb{R}$ , the target is a quantity.

*What credit line should this customer receive?*

The learning machinery is the same; only the shape of  $Y$  — and therefore the loss — changes.

# The Model

**Linear regression** predicts the target as a weighted sum of the features:

$$h_{\mathbf{w}}(\mathbf{x}) = w_0 + w_1x_1 + \dots + w_px_p = \mathbf{w}^\top \mathbf{x}$$

with  $x_0 = 1$  absorbing the intercept.

The hypothesis space is  $\mathbb{R}^{p+1}$ : one hypothesis per weight vector.

# The Objective: Residual Sum of Squares

$$\hat{\mathbf{w}} = \arg \min_{\mathbf{w} \in \mathbb{R}^{p+1}} \text{RSS}(\mathbf{w}), \quad \text{RSS}(\mathbf{w}) = \sum_{i=1}^n (y_i - \mathbf{w}^\top \mathbf{x}_i)^2$$

RSS measures **effectiveness** — how good the predictions are. Do not confuse it with **efficiency**, which is about runtime and computational effort.

# One Feature: The Closed Form

For a single feature the minimiser can be written down directly:

## Slope

$$\hat{w}_1 = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sum_i (x_i - \bar{x})^2}$$

The covariance of  $x$  and  $y$  over the variance of  $x$ .

## Intercept

$$\hat{w}_0 = \bar{y} - \hat{w}_1 \bar{x}$$

The fitted line passes through the point of means.

# How Much Variance Is Explained?

$$\text{TSS} = \text{ESS} + \text{RSS}, \quad R^2 = \frac{\text{ESS}}{\text{TSS}} = 1 - \frac{\text{RSS}}{\text{TSS}}$$

where TSS is the total, ESS the explained and RSS the residual sum of squares (the split holds for a least-squares fit with an intercept).

$R^2$  is the fraction of the target's variation the model accounts for: 0 means no better than predicting the mean, 1 means an exact fit.

**Training**  $R^2$  never decreases when a feature is added, however useless the feature. A high training  $R^2$  is not evidence that a model generalises — test  $R^2$  and adjusted  $R^2$  can and do fall.

# Higher Dimensions

With  $p$  features the same objective in matrix form,

$$\hat{\mathbf{w}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$$

gives the optimum in one step whenever  $\mathbf{X}^\top \mathbf{X}$  is invertible.

Invertibility fails when features are linearly dependent or when  $p > n$  — more unknowns than equations. Both are common in practice; regularisation is the standard remedy.

# Takeaway

Linear regression fits a weighted sum by minimising squared residuals. Its solution is built from **covariance over variance**, and  $R^2$  reports how much variation it explains — not whether it will generalise.

# Linear in the Parameters, Not in the Features

A model is **linear** when it is linear in the **weights**  $\mathbf{w}$ . The features may be arbitrary transformations of the input.

## Plain

$$h(\mathbf{x}) = w_0 + w_1x$$

a straight line

## Polynomial

$$h(\mathbf{x}) = w_0 + w_1x + w_2x^2$$

a curve — still a *linear model*

## Basis functions

$$h(\mathbf{x}) = \sum_j w_j \phi_j(\mathbf{x})$$

any fixed feature map  $\phi$

What *linear* forbids: a transformation that has to be learned from the data. The feature map is fixed before fitting; only the weights are learned.

# Consequences

## The good

The closed-form solution and all statistical guarantees carry over unchanged.

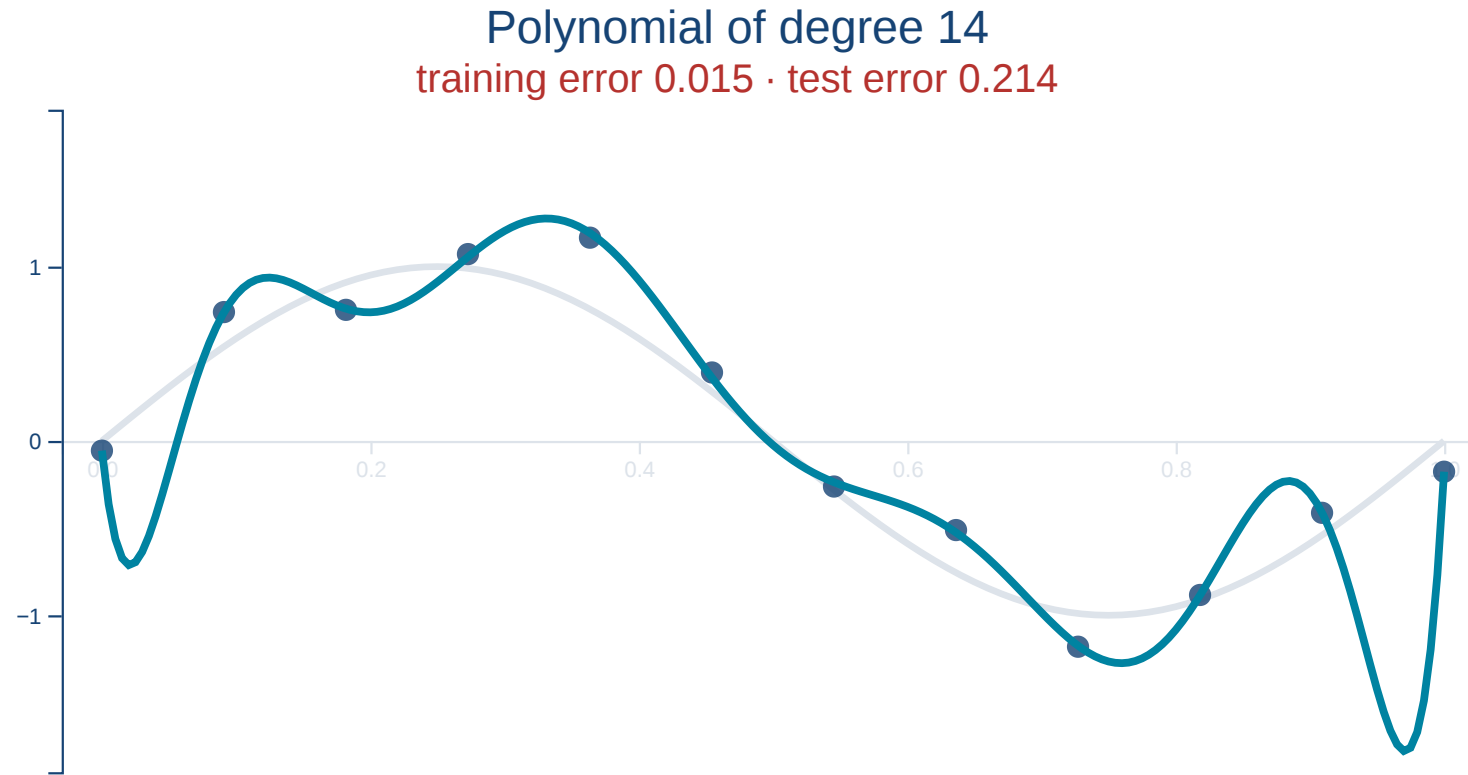
## The catch

The feature map is a **modelling decision**. Choose too rich a basis and the model overfits; too poor and it underfits.

Increasing the polynomial degree turns a dial on the bias–variance axis.

# Turning the Capacity Dial

Linear in the parameters, any basis in the features: a polynomial of degree  $k$  is still "linear regression" — and  $k$  is a dial.



# The Question Nobody Asks

We minimised the *squared* residuals. Why squares — why not absolute deviations, or fourth powers?

# Geometric View

The predictions  $\mathbf{X}\mathbf{w}$  live in the **column space** of  $\mathbf{X}$  — everything the features can express. The observed  $\mathbf{y}$  generally does not.

Least squares picks the point of that space **closest to  $\mathbf{y}$  in Euclidean distance**: the orthogonal projection. The residual is perpendicular to every feature.

## Consequence

The residual carries no remaining linear information about the features.

# The Gauss–Markov Theorem

**Gauss–Markov (informal).** Among all estimators that are

- **linear** in the observed outputs  $(y_1, \dots, y_n)$ , and
- **unbiased**,

the least-squares estimator has the **lowest variance**. It is the *Best Linear Unbiased Estimator* — BLUE.

## Best

lowest variance among the candidates

## Linear

a linear function of the observations

## Unbiased

right on average over repeated samples

# What This Buys, and What It Costs

## The payoff

Minimising RSS yields optimal statistical properties under **mild assumptions** — no normality needed, only a correctly specified linear model, linearly independent features, and zero-mean, uncorrelated errors of constant variance.

## The price

Alternatives — absolute error, Huber loss — are **more robust to outliers** but give up the BLUE guarantee.

Squared error is not an arbitrary convenience: it is the loss whose minimiser is the best linear unbiased estimator. Departing from it is legitimate — but it is a trade, and you should be able to name what you traded.

# Two Ways to the Same Minimum

## Direct (closed form)

$$\hat{\mathbf{w}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$$

Exact, one step. Requires forming and inverting a  $(p+1) \times (p+1)$  matrix — costly for large  $p$ , impossible when it is singular.

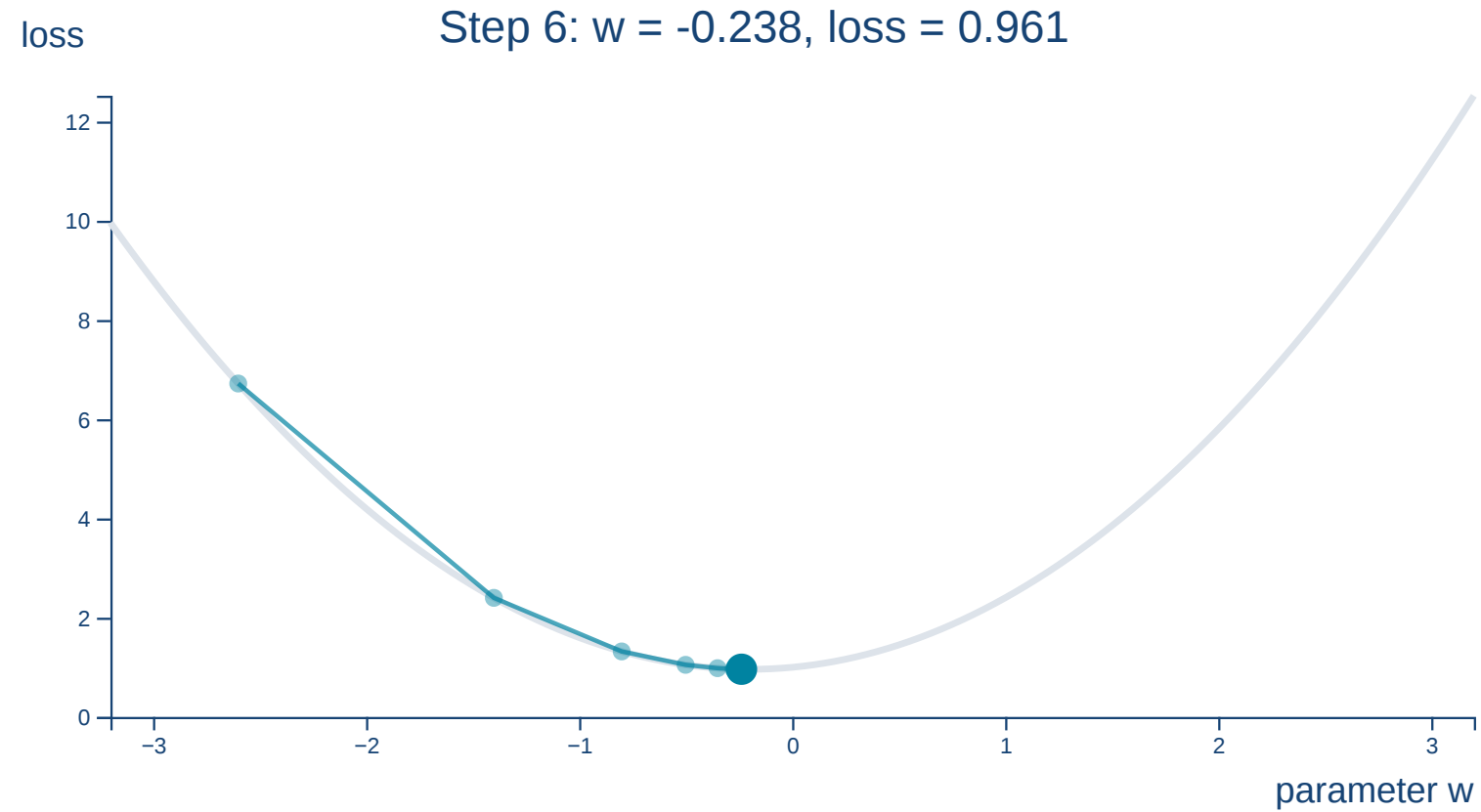
## Iterative (gradient descent)

$$\mathbf{w} \leftarrow \mathbf{w} - \eta \nabla \text{RSS}(\mathbf{w})$$

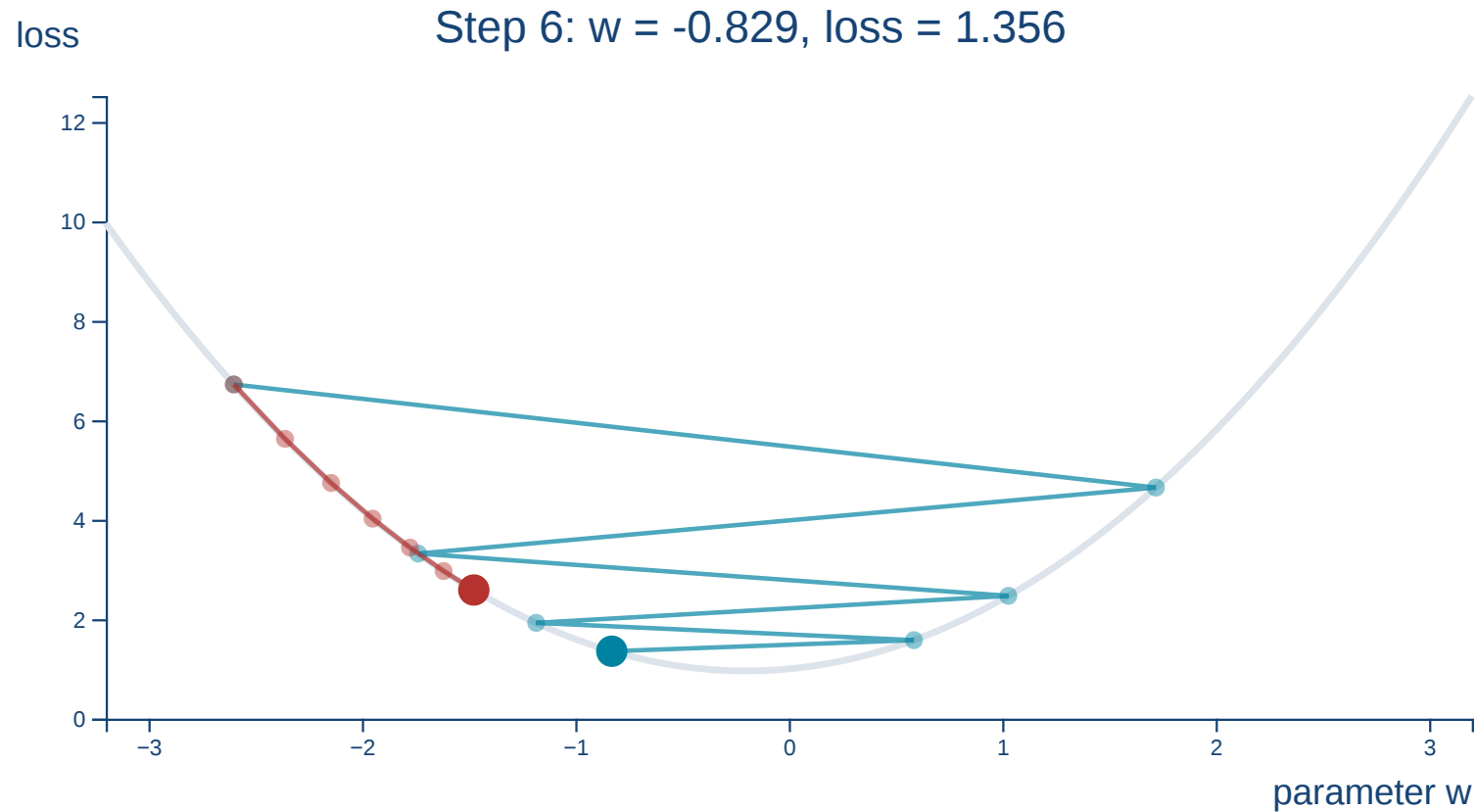
Approximate, many steps. Scales to large  $p$  and  $n$ , works when the direct solution does not exist, and extends to models with no closed form at all.

The iterative rule is exactly the **LMS algorithm**: its update  $w_i \leftarrow w_i + \eta e x_i$  is a gradient step on RSS, taken per example instead of per dataset (the factor 2 is absorbed into  $\eta$ ).

# Watching Gradient Descent Work



# The Learning Rate Decides Everything



Teal:  $\eta = 0.9$  — overshoots and oscillates across the minimum. Red:  $\eta = 0.05$  — stable, but slow: after six steps it has covered about half the distance.



The learning rate is a **hyperparameter**: it is chosen, not learned, and both failure modes are silent — one looks unstable, the other looks converged.

# Takeaway

Use the **closed form** when the matrix is small and well-conditioned; use **gradient descent** when it is not — and for most models beyond linear regression, no closed form exists at all.